

Project Overview

Topic: Neurosymbolic Reasoning

Huỳnh Gia Bảo, Nguyễn Ngọc Cảnh, Nguyễn Đức Trí,
Phan Huỳnh Minh Khôi, Trần Lý Nhật Hào

Faculty of Information Technology
University of Science - VNUHCM

January 26, 2026

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$	
A	
$\therefore B$	(Syllogism/Modus ponens)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$	
A	
$\therefore B$	(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A
—
 $\therefore B$
(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A
—
 $\therefore B$

(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

Cons

- Brittleness
- Scalability
- Poor Perception

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A

 $\therefore B$

(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

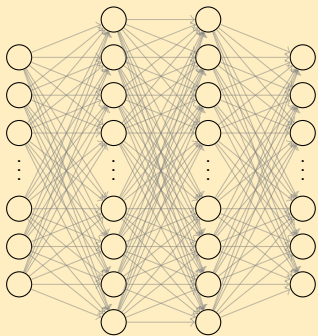
Cons

- Brittleness
- Scalability
- Poor Perception

$\Rightarrow$ **reliable but inflexible.**

Modern AI: Data-driven/Neural Networks

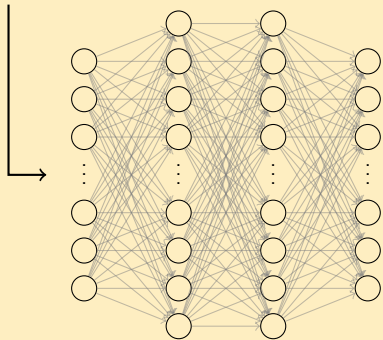
Neural Networks



Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy wibu?”

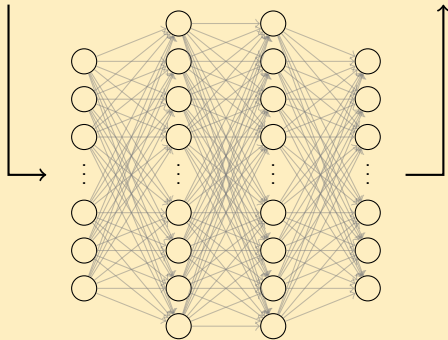


Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy wibu?”

“Yes!”

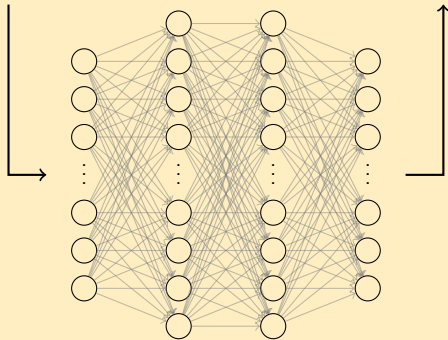


Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy wibu?”

“Yes!”



Pros

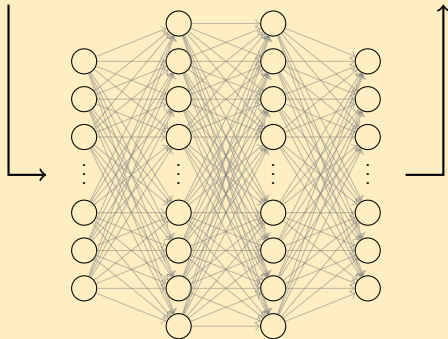
- Generalization and Robustness
- Perception
- Automated Feature Extraction

Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy wibu?”

“Yes!”



Pros

- Generalization and Robustness
- Perception
- Automated Feature Extraction

Cons

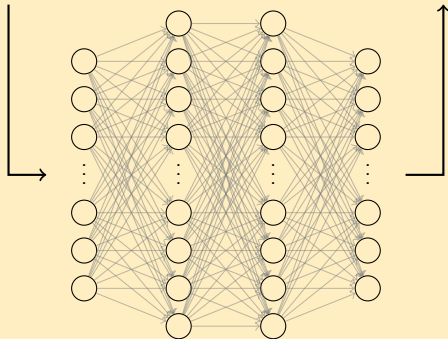
- Lack of Explainability
- Data & Resource Hungry
- Hallucination

Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy wibu?”

“Yes!”



Pros

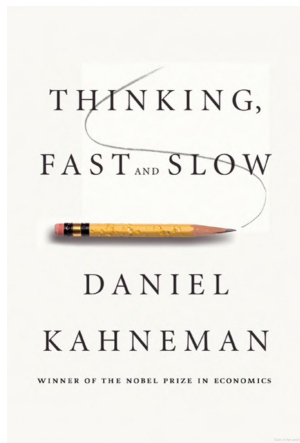
- Generalization and Robustness
- Perception
- Automated Feature Extraction

Cons

- Lack of Explainability
- Data & Resource Hungry
- Hallucination

⇒ **flexible but unreliable.**

Neurosymbolic AI: A Hybrid Approach



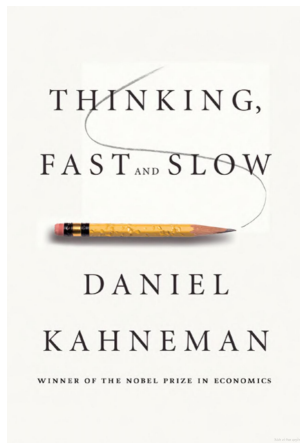
SYSTEM 1 (FAST)

Intuition & Instinct

SYSTEM 2 (SLOW)

Rational thinking

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

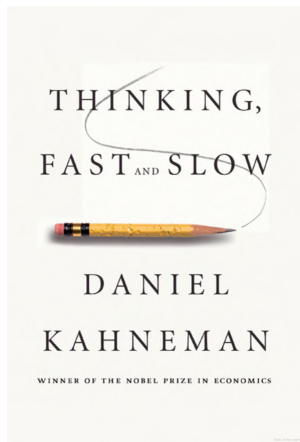
SYSTEM 2 (SLOW)

Rational thinking

||

NEURAL NETWORK

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

||

NEURAL NETWORK

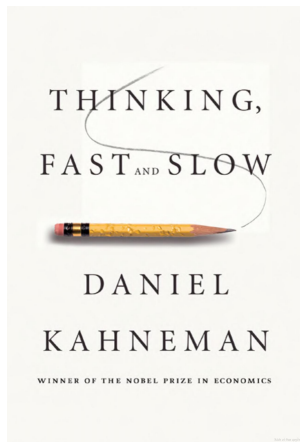
SYSTEM 2 (SLOW)

Rational thinking

||

SYMBOLIC ENGINE

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

||

NEURAL NETWORK

+

SYSTEM 2 (SLOW)

Rational thinking

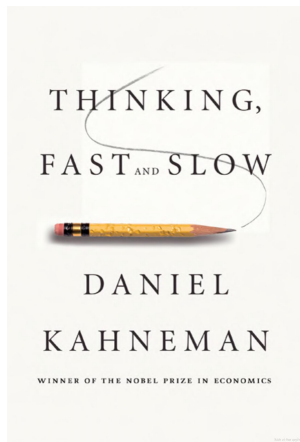
||

SYMBOLIC ENGINE

=

**HUMAN
INTELLIGENCE**

Neurosymbolic AI: A Hybrid Approach



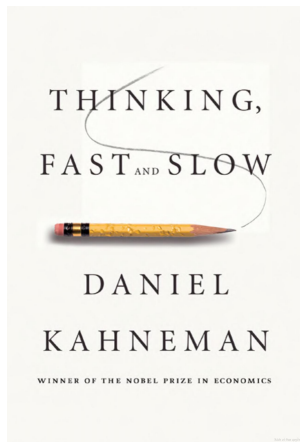
SYSTEM 1 (FAST) + **SYSTEM 2 (SLOW)** = **HUMAN INTELLIGENCE**
Intuition & Instinct Rational thinking

||

||

NEURAL NETWORK + **SYMBOLIC ENGINE** = **NEUROSYMBOLIC AI**

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST) + **SYSTEM 2 (SLOW)** = **HUMAN INTELLIGENCE**
Intuition & Instinct Rational thinking

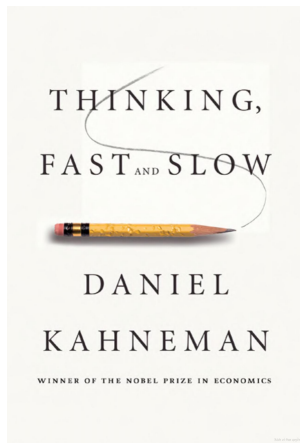
||

||

||

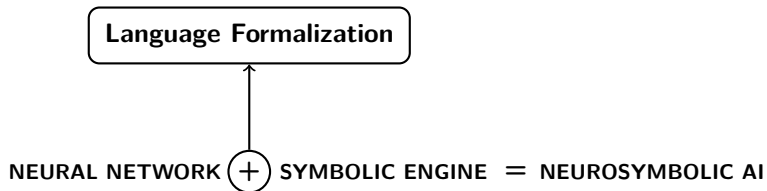
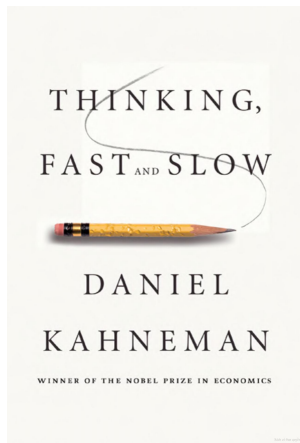
NEURAL NETWORK + **SYMBOLIC ENGINE** = **NEUROSYMBOLIC AI**

Neurosymbolic AI: A Hybrid Approach



NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

Neurosymbolic AI: A Hybrid Approach



Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement**
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Language Formalization

Input/Output

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data**
- 4 Related Work
- 5 Challenges & Solutions

Data Source

The data used consists of the training sets of the datasets listed in the table below.

Source	Volume	Description
FOLIO	1,000 samples	With FOL label , with true/false/uncertain label.
ProofWriter	41,400 samples	Without FOL label , with true/false label, with proof.
RuleTaker	480,000 samples	Without FOL label , with entailment/not entailment label.
ReClor	4,637 samples	Without FOL label , in multiple-choice format.
LogiQA	7,380 samples	Without FOL label , in multiple choice format.

Table: Preliminary statistics of data sources.

Data Pipeline

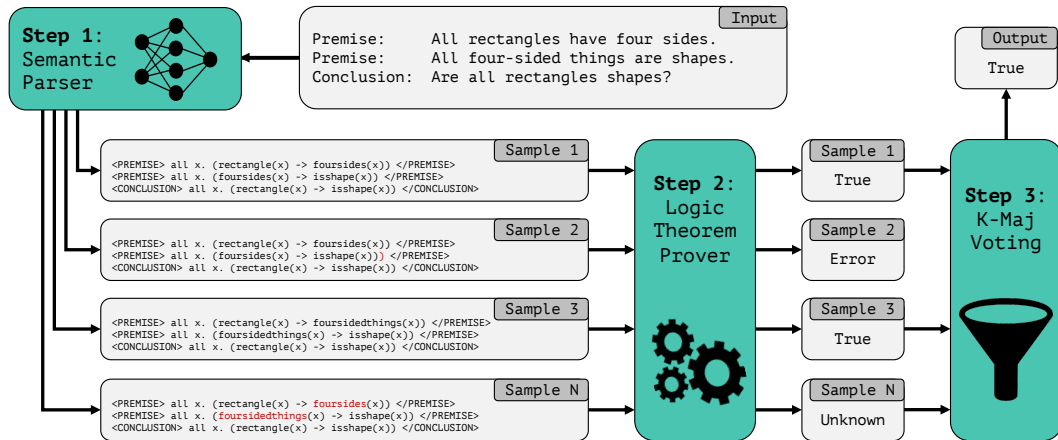


Figure: Mình không hẳn là dùng pipeline này, để tạm đây mai sửa

Data Pipeline

(Viết chi tiết aabcxyz về data pipeline, giai đoạn 1 dùng SFT, giai đoạn 2 dùng DPO)

Supervised Fine-Tuning (SFT):

$$L_{\text{SFT}}(\theta) = - \sum \log \mathbb{P}(y_{\text{FOL}} \mid x_{\text{NL}}).$$

Direct Preference Optimization (DPO):

$$L_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right].$$

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work**
- 5 Challenges & Solutions

Related Work

Liệt kê ngắn gọn đóng góp của các paper liên quan

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions**

Anticipated Challenges & Proposed Solutions

Challenge X

XXX

Solution Y

YYY

Thank you!